Summary in Graph

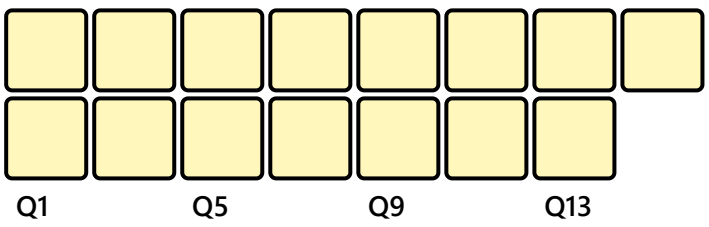
Exam Summary (GO Classes GATE CS | Digital Logic | Number System | Weekly Quiz 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (22) out of 215 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Digital Logic

The number 233_{10} is equal to the following:

A. 10101001_2

B. 11001001_2

C. 11111001_2

D. 11101001_2

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The number 233_8 is equal to the following:

A. 150_{10}

B. 164_{10}

C. 155_{10}

D. 159_{10}

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

What is the magnitude of the 10-bit $2's$ complement number 1011100010_2?

A. 286_{10}

B. -739_{10}

C. 738_{10}

D. -286_{10}

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Digital Logic

Which representation schemes allow performing the 9's complement of decimal digits by simply taking the one's complement of their binary representations?

A. 2421 code

B. the 2's complement number system

C. XS-3 code (Excess-3 Code)

D. 8421 code

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Digital Logic

The string $ABBA$ may represent:

A. A number in base 16

B. A number in base 10

C. A number in base 14

D. A 70s pop group

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Express the unsigned fixed-point binary 0011110010_2 as a decimal, assuming the format $bbbbb.bbbbb_2$:

A. 7.06250

B. 6.56250

C. -7.56250

D. 7.56250

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

The hexadecimal number 67565_{16} has a value:

A. $= 1472545_8$

B. $< 67565_{10}$

C. $< F0000_{16}$

D. $\leq 1000000_{10}$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

The sum $110101_2 + 101_2$ is equivalent to:

A. None of the others

B. 111010_{10}

C. $= 65_8 + 5_8$

D. 00111010_2

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

The binary product $110101_2 \times 101_2$ is equivalent to:

A. None of the others

B. $110101_2 + 11010100_2$

C. $110101_2 + 1101010_2$

D. $5_{10} \times 65_8$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 01sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The binary number 11010.110001 is equivalent to:

A. 1.110110001×2^6

B. 1.110110001×2^4

C. 0.11010110001×2^5

D. 1.110110001×2^5

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

What are the decimal values of the binary number (8 bits) 10011001, considering the unsigned representation and the two's complement representation, respectively?

A. $153, -102$

B. $100, -101$

C. $153, -103$

D. $129, 103$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

A number in 4-bit 2s complement signed representation is $a_3a_2a_1a_0$. The same number, when stored using 8 bits, will appear like:

A. $0000\ a_3\ a_2\ a_1\ a_0$

B. $a_3' \; a_3' \; a_3' \; a_3' \; a_3 \; a_2 \; a_1 \; a_0$

C. $a_3 \; a_2 \; a_1 \; a_0 \; a_3 \; a_2 \; a_1 \; a_0$

D. $a_3 \; a_3 \; a_3 \; a_3 \; a_3 \; a_2 \; a_1 \; a_0$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The octal number 326.4 is equivalent to

A. $(214.2)_{10}$ and $(D6.8)_{16}$

B. $(212.5)_{10}$ and $(D6.8)_{16}$

C. $(214.5)_{10}$ and $(D6.8)_{16}$

D. $(214.5)_{10}$ and $(D6.4)_{16}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Given $(135)_x + (144)_x = (323)_x$ What is the value of base x ?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

How many numbers in the range $0, 1, \ldots, 1365$ have exactly four 1's in their binary representation? (Hint: 1365_{10} is 10101010101_2, that is,

$$1365 = 2^{10} + 2^8 + 2^6 + 2^4 + 2^2 + 2^0.)$$

In the following, the binomial coefficient $\binom{n}{k}$ counts the number of k-element subsets of an n-element set.

A. $\binom{6}{4}$

B. $\binom{10}{4}$

C. $\binom{10}{4} + \binom{8}{3} + \binom{6}{2} + \binom{5}{1}$

D. $\binom{11}{4} + \binom{9}{3} + \binom{7}{2} + \binom{5}{1}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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